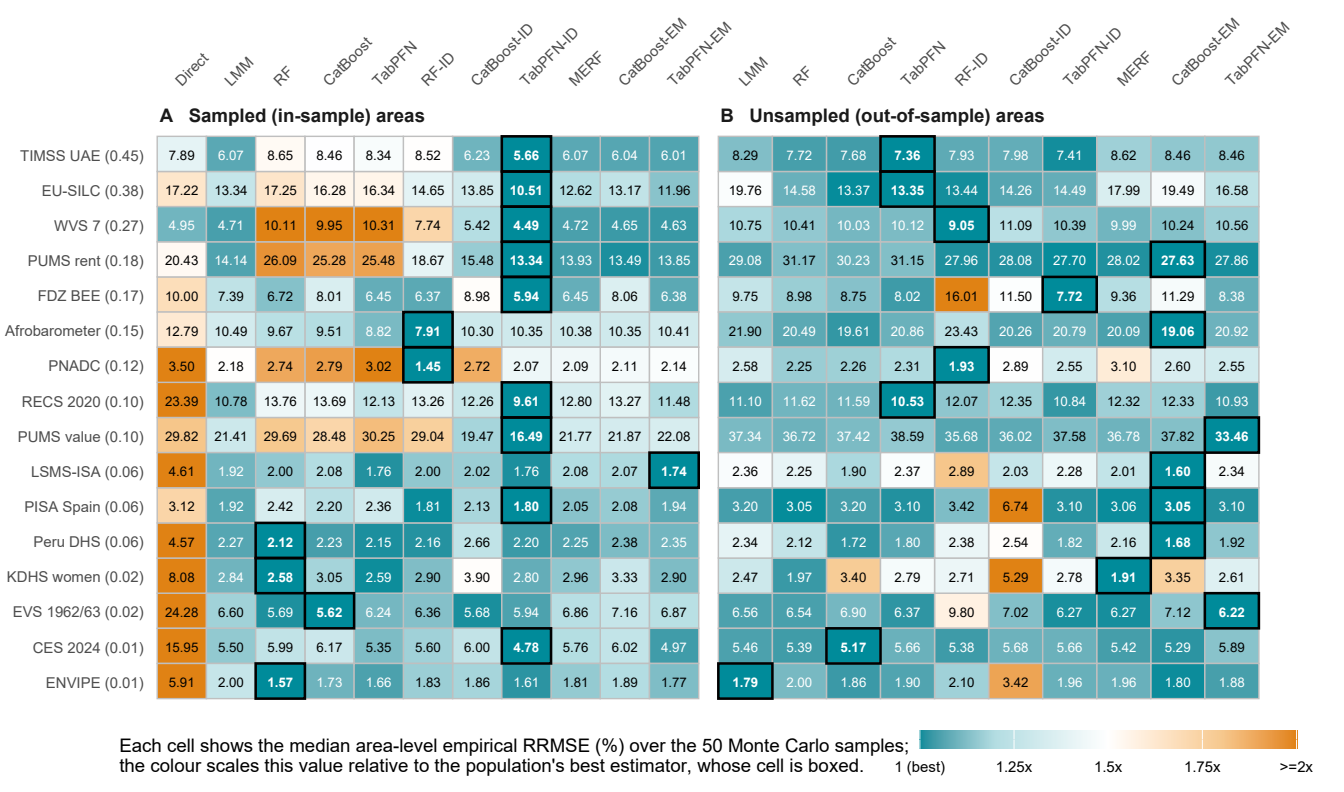
Direct
LMM
RF
CatBoost
TabPFN
RF-ID
CatBoost-ID
TabPFN-ID
MERF
CatBoost-EM
TabPFN-EM
A Sampled (in-sample) areas
B Unsampled (out-of-sample) areas
TIMSS UAE (0.45) 7.89 6.07 8.65 8.46 8.34 8.52 6.23 5.66 6.07 6.04 6.01
EU-SILC (0.38) 17.22 13.34 17.25 16.28 16.34 14.65 13.85 10.51 12.62 13.17 11.96
WVS 7 (0.27) 4.95 4.71 10.11 9.95 10.31 7.74 5.42 4.49 4.72 4.65 4.63
PUMS rent (0.18) 20.43 14.14 26.09 25.28 25.48 18.67 15.48 13.34 13.93 13.49 13.85
FDZ BEE (0.17) 10.00 7.39 6.72 8.01 6.45 6.37 8.98 5.94 6.45 8.06 6.38
Afrobarometer (0.15) 12.79 10.49 9.67 9.51 8.82 7.91 10.30 10.35 10.38 10.35 10.41
PNADC (0.12) 3.50 2.18 2.74 2.79 3.02 1.45 2.72 2.07 2.09 2.11 2.14
RECS 2020 (0.10) 23.39 10.78 13.76 13.69 12.13 13.26 12.26 9.61 12.80 13.27 11.48
PUMS value (0.10) 29.82 21.41 29.69 28.48 30.25 29.04 19.47 16.49 21.77 21.87 22.08
LSMS-ISA (0.06) 4.61 1.92 2.00 2.08 1.76 2.00 2.02 1.76 2.08 2.07 1.74
PISA Spain (0.06) 3.12 1.92 2.42 2.20 2.36 1.81 2.13 1.80 2.05 2.08 1.94
Peru DHS (0.06) 4.57 2.27 2.12 2.23 2.15 2.16 2.66 2.20 2.25 2.38 2.35
KDHS women (0.02) 8.08 2.84 2.58 3.05 2.59 2.90 3.90 2.80 2.96 3.33 2.90
EVS 1962/63 (0.02) 24.28 6.60 5.69 5.62 6.24 6.36 5.68 5.94 6.86 7.16 6.87
CES 2024 (0.01) 15.95 5.50 5.99 6.17 5.35 5.60 6.00 4.78 5.76 6.02 4.97
ENVIPE (0.01) 5.91 2.00 1.57 1.73 1.66 1.83 1.86 1.61 1.81 1.89 1.77
LMM RF CatBoost TabPFN RF-ID CatBoost-ID TabPFN-ID MERF CatBoost-EM TabPFN-EM
TIMSS UAE (0.45) 8.29 7.72 7.68 7.36 7.93 7.98 7.41 8.62 8.46 8.46
EU-SILC (0.38) 19.76 14.58 13.37 13.35 13.44 14.26 14.49 17.99 19.49 16.58
WVS 7 (0.27) 10.75 10.41 10.03 10.12 9.05 11.09 10.39 9.99 10.24 10.56
PUMS rent (0.18) 29.08 31.17 30.23 31.15 27.96 28.08 27.70 28.02 27.63 27.86
FDZ BEE (0.17) 9.75 8.98 8.75 8.02 16.01 11.50 7.72 9.36 11.29 8.38
Afrobarometer (0.15) 21.90 20.49 19.61 20.86 23.43 20.26 20.79 20.09 19.06 20.92
PNADC (0.12) 2.58 2.25 2.26 2.31 1.93 2.89 2.55 3.10 2.60 2.55
RECS 2020 (0.10) 11.10 11.62 11.59 10.53 12.07 12.35 10.84 12.32 12.33 10.93
PUMS value (0.10) 37.34 36.72 37.42 38.59 35.68 36.02 37.58 36.78 37.82 33.46
LSMS-ISA (0.06) 2.36 2.25 1.90 2.37 2.89 2.03 2.28 2.01 1.60 2.34
PISA Spain (0.06) 3.20 3.05 3.20 3.10 3.42 6.74 3.10 3.06 3.05 3.10
Peru DHS (0.06) 2.34 2.12 1.72 1.80 2.38 2.54 1.82 2.16 1.68 1.92
KDHS women (0.02) 2.47 1.97 3.40 2.79 2.71 5.29 2.78 1.91 3.35 2.61
EVS 1962/63 (0.02) 6.56 6.54 6.90 6.37 9.80 7.02 6.27 6.27 7.12 6.22
CES 2024 (0.01) 5.46 5.39 5.17 5.66 5.38 5.68 5.66 5.42 5.29 5.89
ENVIPE (0.01) 1.79 2.00 1.86 1.90 2.10 3.42 1.96 1.96 1.80 1.88
Each cell shows the median area-level empirical RRMSE (%) over the 50 Monte Carlo samples;
the colour scales this value relative to the population's best estimator, whose cell is boxed.
1 (best)
1.25x
1.5x
1.75x
>=2x